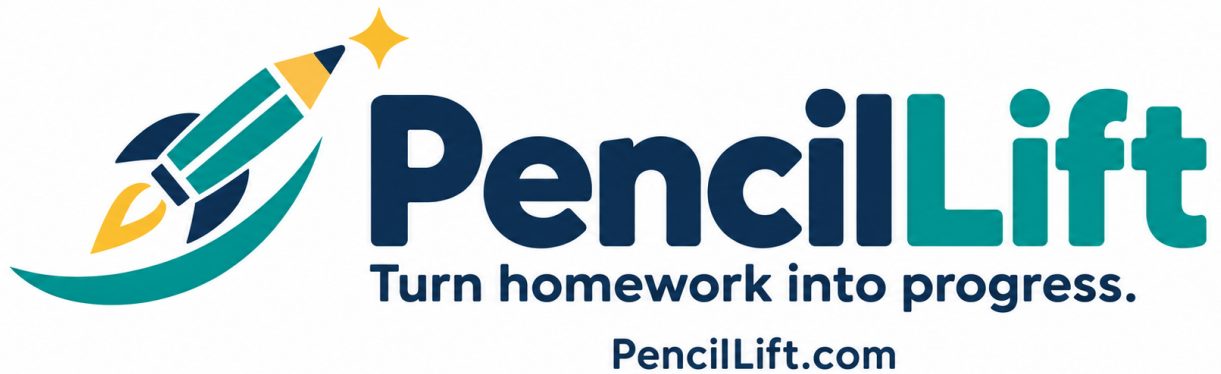


PencilLift

Build brief, AI recommendation and family economics



A homework coach that helps children learn the method, gives parents the full picture, and turns practice into progress.

This package preserves the approved pencil-rocket logo and documents the AI-centered app: protected parent/child access, scan checking, personalized practice, reviews, rewards, subscriptions and native store releases. Includes parent sponsorships and conditional Amazon affiliate support.

Recommended starting route: GPT-5.6 Terra for scans and private checking; GPT-6 Astra for original child teaching and difficult questions; GPT-5.6 Luna for adult summaries.

Approved monthly price: \$39.99 for the first child plus \$9.99 for each additional child. Two children: \$49.98; three: \$59.97; four: \$69.96. One parent account, separate child profiles. Costs and profitability still require pilot validation.

Specification only: the app has not been built, benchmarked or submitted. A complete prompt still requires implementation, internal test/fix cycles, owner account setup and external store review. All usage and labor estimates need pilot validation.

What the software must do

Area	Required behavior
Parent access	Verified adult account, expiring child-pairing code and protected parent unlock. Parents see answers, worked explanations, skill patterns, rewards, subscriptions and resources.
Child access	No child email. A child-scoped session shows correct / try again, method hints and analogous examples. The original answer key never reaches the child client or its logs/cache.
Scan and check	Camera, photo or PDF upload; crop/retake; handwriting and diagram extraction; independent verification; uncertainty and unreadable states; retry/resume without duplicate billing.
Skill progress	Track subject and skill evidence across independent and assisted attempts. Identify patterns without medical diagnoses or unsupported claims of mastery. Parents can correct grading.
Daily practice	Five extra-credit questions every day, weighted toward weak skills, spaced review and confidence building. AI adapts sets with reviewed grounding and safe fallbacks within plan limits.
Thursday review	Prepare reviews at 4 p.m. in the family's local time zone, even with the app closed. Default eight questions per active subject, mostly weak skills, ready for Friday tests.
Rewards	Deterministic points, an auditable ledger, parent-defined goals and parent approval of redemption. Parents deliver money or goods outside the app.
Parent resources	Explain why a reviewed workbook or tutoring resource matches a skill. Amazon links stay in the adult area. No invented listings, prices or child-data sharing.
Mobile + web	Native iOS and Android apps plus a parent web portal. Real subscriptions, purchase restore, account deletion, notifications, accessibility and error/offline states.

Proposed initial scope: US English, kindergarten–grade 8, initial capacity tiers for one to four paid children and two guardians. Actual curriculum coverage must be declared and reviewed; unsupported work is routed to parent review. Profile capacity is distinct from paid usage limits.

Give the coding agent a complete contract

The build stack

React Native with Expo and Expo Router for iOS/Android; a React parent web portal; a TypeScript API and durable worker queue; Supabase Postgres, Auth and private storage; RevenueCat for native subscription entitlements; OpenAI through server-side requests. Native store purchases use StoreKit and Google Play Billing.

The approved brand

Keep PencilLift and the upward pencil-rocket from the supplied reference. Use navy #17324D, teal #008D87, gold #FFB84D and white. Preserve “Turn homework into progress.” and PencilLift.com. The brand guide supplies an exact reproduction prompt and production icon requirements. The reference board is not itself a ready-to-upload app icon or vector file.

The documents that drive implementation

File	Purpose
PencilLift_Master_Build_Prompt.md	Complete agent instructions: roles, screens, schemas, API contracts, AI stages, schedules, payments, privacy and release evidence.
PencilLift_Acceptance_Checklist.md	Observable pass/fail requirements for user flows, isolation, answer protection, scheduling, duplicate events, purchases and deletion.
Owner_Setup_Checklist.md + config.example.env	Required owner accounts, credentials, consent setup, store settings and safe configuration placeholders.
brand/BRAND_GUIDE.md + logo PNG	Approved source and requirements for faithful production assets.
Child pricing + cost report + calculator + JSON	Approved base/sibling prices, child-slot billing, dated cost assumptions and editable per-family economics.
Monetization plan + revenue planner	Parent sponsor placements, affiliate eligibility, disclosures, revenue scenarios and section 17 of the master prompt.

Unzip the package into the project. Tell Claude Code or Codex to read the full master prompt and all referenced files, implement every requirement, then test and fix until the completion gates pass.

The agent must maintain a requirement-to-code-to-test matrix, use synthetic test data, deliver runnable source and migrations, and report each integration as implemented, verified, or blocked by a named external dependency. It must never claim a mock integration or unsigned build is a live store release.

Use GPT-6 where it adds the most value

Model / API ID	Input / 1M	Output / 1M	Planned role
GPT-5.6 Terra gpt-5.6-terra	\$2.00	\$12.00	Read homework images, private grading and verification
GPT-6 Astra gpt-6-astra	\$10.00	\$50.00	Original child coaching, custom practice, Thursday reviews and hard cases
GPT-5.6 Luna gpt-5.6-luna	\$0.20	\$1.20	Adult summaries and reviewed resource classification

These are standard short-context API rates, not ChatGPT subscription prices. Both Terra and Astra accept images. Input budgets include image tokens; billable output includes reasoning. The plan assumes no caching or Batch discount. Long context, additional tools and special processing tiers can change charges.

[Official OpenAI model pricing](#)

Check quality before trusting the route

Benchmark at least 200 reviewed, representative questions and scans across intended subjects, grades and handwriting conditions. Measure correctness, safe abstention, original-answer leakage, latency and tokens by feature. The master prompt defines proposed acceptance thresholds; none have been demonstrated yet. Deterministic math and vetted answer templates should handle eligible routine tasks.

Two production prerequisites for children

Verified parental consent: implement an appropriate provider-backed process, notices, withdrawal, access and deletion. A parent PIN, checkbox or purchase gate alone is not a substitute for the required consent method.

OpenAI Zero Data Retention: obtain actual eligible account approval and verify the model/endpoint setup before sending under-13 personal data. Setting response storage to false alone does not establish ZDR. The app must fail closed for such live processing until prerequisites are satisfied.

[OpenAI under-18 guidance and child-data requirements](#)

[OpenAI data controls and ZDR eligibility](#)

[FTC COPPA compliance guidance](#)

The architecture keeps answer keys on the server, isolates every family and child, uses short-lived private file access, prevents homework from becoming model instructions, and excludes child data from third-party advertising. Store child-audience rules must also be checked against the actual AI and SDK data flows.

Original two-child runtime: \$17.26

Original scenario: 80 pages/month; five daily questions per child for 30 days; four subjects × eight Thursday questions across 4.33 weeks; 10 new explanations, 20 coaching turns and eight custom daily sets per child. Other practice uses the bank. Twenty percent of practice responses need AI checking; 5% of pages get an extra Astra check. Re-measure for the revised AI-centered workload.

AI stage	Model	Calls / month	Cost
Vision extraction	Terra	80	\$1.79
Private grading	Terra	80	\$2.32
Original child explanations	Astra	20	\$1.70
Child follow-up coaching	Astra	40	\$1.48
Custom daily practice sets	Astra	16	\$1.36
Thursday review bundles	Astra	8.67	\$1.95
Independent verification	Terra	164.67	\$1.65
Semantic practice checking	Terra	115.47	\$0.69
Difficult-case escalation	Astra	4	\$0.56
Adult weekly summary/catalog match	Luna	4.33	\$0.01
AI usage subtotal			\$13.51
25% uncertainty reserve			\$3.38
AI budget			\$16.89

Add \$0.25 marginal infrastructure allowance plus \$0.12 shared allocation at 1,000 families to reach **\$17.26 system runtime**. The shared business baseline is \$118.92/month. Its starting plans are not a guaranteed capacity level.

All-GPT-6 comparison: using Astra at every modeled stage with the same counts raises runtime to about **\$45.33/family/month**. The hybrid is about 62% lower in this estimate. This is a cost comparison, not evidence of equal quality.

Formula: $\text{calls} \times (\text{input tokens} \times \text{input rate} + \text{billable output tokens} \times \text{output rate}) / 1,000,000$. Packet verification is included. Dense scans, essays, long reasoning, a missing reusable content bank, repeated failures and more custom coaching can increase costs substantially.

One base price, then add children

\$39.99/month for the first child, plus \$9.99/month for each additional child on the same parent account. Monthly USD totals below are owner-approved; actual store checkout must match the available catalog, currency and applicable tax rules.

Paid children	Monthly price	Modeled runtime	With fees, support + consent	Contribution before overhead
1	\$39.99	\$11.43	\$20.25	\$19.74
2	\$49.98	\$22.48	\$32.90	\$17.08
3	\$59.97	\$33.53	\$45.54	\$14.43
4	\$69.96	\$44.58	\$58.19	\$11.77

Unmeasured planning scenario: 40 homework pages and 30 freshly generated daily sets per child/month, plus 10 original explanations, 20 coaching turns and the earlier Thursday/verifier assumptions. Retains the dated token estimates and 25% AI reserve. Shared costs are allocated across 1,000 paying families; commercial revenue is zero.

Fees assume 15% store commission plus 1% RevenueCat when applicable. Support is \$2/family and consent is a \$5 onboarding allowance over 12 months. These are not quotes. Actual per-child verification/support and heavy usage may cost more.

The two-child price is \$49.98/month. This scenario leaves about \$17.08 before maintenance, marketing, refunds, tax effects, development and owner compensation. It does not establish net profit.

How parents add or remove a child

A draft profile creates no charge. Assign an unused paid slot, or show the upgraded recurring total and obtain parent/store confirmation. Activate new capacity only after verified purchase. Each child keeps separate learning history, practice, rewards and usage. Downgrades follow the store-confirmed date and preserve parent-readable history.

Each \$9.99 additional child brings about \$8.39 after the assumed percentage fees, versus about \$11.05 modeled extra runtime here. The base subscription absorbs the difference. Validate sibling margins in a pilot before claiming profitability.

Store catalogs must support the exact totals. Apple uses predefined price points; do not silently round \$49.98, \$59.97 or \$69.96. The builder must report unavailable prices before activating products. Use capacity-tier subscriptions and verified slot mappings.

[Apple subscription groups and capacity-tier implementation](#)

[Apple subscription price points](#)

Shared baseline and pricing levers

Service / allowance	USD / month	Basis
Supabase Pro	\$25.00	Published starting price
Cloudflare Workers	\$5.00	Published paid minimum; additional queue/CPU usage possible
Expo EAS Starter	\$19.00	Published base; build/update overages possible
Resend Pro	\$20.00	Published base; 50,000 emails/month
Monitoring allowance	\$20.00	Planning allowance, not a vendor quote
Extra backups/staging allowance	\$20.00	Planning allowance, not a vendor quote
Apple membership monthly allocation	\$8.25	Published \$99/year
Domain renewal monthly allowance	\$1.67	Planning allowance; actual purchased-domain renewal unknown
Shared baseline	\$118.92	Allocated across active families

These are starting plans and budget allowances, not an infrastructure capacity guarantee. Additional compute, queues, storage, egress, builds and emails can increase the bill. The domain has already been purchased; its renewal is a future expense.

Measure the expensive parts first

At 1,000 two-child families, the 30-fresh-daily-set scenario models about \$22,482/month system runtime including the AI reserve. Track custom coaching, scan density, retry rates and long reasoning separately. Trial accounts cost money before they convert. Avoid unlimited usage until actual distributions and plan limits support it.

The approved first-child price is \$39.99, with each additional child \$9.99/month. Preserve AI and eligible commercial modules. Prototype usage is 40 pages per paid child/month plus bounded coaching and practice; validate final advertised limits. Model fees and costs by paid child count, with zero commercial revenue as the baseline.

Optional US web Stripe billing is a separate channel: the published 2.9% + \$0.30 card rate plus 0.7% Billing would be about \$1.74 at \$39.99. It is not added to the same native-store transaction. Regional store rules and RevenueCat tracking still need review.

Sources: [Supabase pricing](#)

[Cloudflare Workers pricing](#)

[Expo build pricing](#)

[Resend email pricing](#)

[Stripe Billing pricing](#)

The remaining costs and release steps

Startup item	Planning amount
Apple Developer Program	\$99/year
Google Play Console	\$25 one time
Development AI tools and experiments	\$200-\$1,500
Child-privacy/legal/store-policy review	\$2,000-\$8,000
Independent security/engineering/release review	\$3,000-\$12,000
Consent-provider setup (quote required)	\$0-\$2,000+
Devices, accessibility and educator testing	\$500-\$3,000
Initial original/licensed content-bank QA	\$500-\$3,000

The non-store planning allowances total **\$6,200-\$29,500+**, excluding a full contracted build and owner labor. They are optional budgeting estimates, not vendor quotes. The first-year store enrollments total \$124 before tax. Apple is also allocated monthly in runtime; do not charge it twice in a cash budget.

Budget a separate **\$500-\$3,000+/month** initial maintenance/content reserve if useful for planning. Marketing, accounting, insurance, taxes, refunds, staff expansion and any quoted consent/ZDR minimums remain additional. Families pay their own reward money and Amazon purchases.

Before store submission

The owner supplies developer accounts, business identity, signing access, billing/tax agreements, DNS, privacy/support contacts, AI account access and consent-provider credentials. The builder delivers signed test builds when access allows, real sandbox purchase/restore tests, live support/privacy/deletion pages, reviewer accounts and accurate child-audience/data disclosures.

Applicable new Google personal accounts require a closed test with at least 12 testers for 14 continuous days. Apple and Google make the final review decisions. Parent gating alone does not resolve their child-audience data requirements. Default Amazon links remain ordinary adult-only links; affiliate eligibility must be established separately.

[Apple enrollment / \\$99 annual membership](#)

[Google developer registration / \\$25 fee](#)

[Google personal-account testing requirements](#)

Full dated citations, fee qualifications and detailed release requirements are in PencilLift_Costs_and_Launch.md and the master build prompt. Start with START_HERE.md.

Parent ads and affiliate support

Keep AI central to homework checking, personalized explanations, daily practice, Thursday reviews and learning insights. Add commercial support in the adult experience, with actual eligibility and earnings kept separate from technical implementation.

Module	Required behavior
Sponsor cards	First-party-hosted, human-reviewed education ads in permitted parent surfaces. One card per screen; dismiss/report controls; schedules, caps, ad-free access and kill switches.
Amazon adapter	Reviewed product IDs, approved link tools, nearby disclosure and free adult access where eligible. Live activation remains blocked until actual property/mobile use is established as eligible.
Child experience	No ads, affiliate links, commercial network requests or shopping incentives. Learning points never depend on ad views, affiliate clicks or purchases.
Owner reporting	Campaign admin and aggregate reports. Separate projected, contracted, recognized and received revenue, reversals and selling/serving costs.

Amazon presents a real eligibility obstacle

Its rules exclude child-directed properties and certain handling of under-threshold child data. A parent PIN, generic Associates account or separate parent page does not resolve that. Build the module, but count no affiliate revenue until the actual use is established as eligible. Parent-only sponsorships also remain subject to app-store commercial-content rules.

Revenue illustration, not a forecast

At 1,000 active families: 40 billable parent impressions each at an assumed \$3 publisher eCPM yield \$120/month. If 20% of families each generate \$25 of qualifying physical-book spend at the current 4.5% category rate, theoretical affiliate income adds \$225. Combined: \$345/month, or \$0.345/family before incremental costs and only if eligible.

A fixed \$500 sponsorship would yield \$0.50/family at that scale before costs; this is an arithmetic scenario, not a signed deal or market quote. Replace network income on that same inventory rather than counting both. No commercial income has been established.

[Amazon eligibility and mobile-app requirements](#)

[Amazon category commission rates](#)

[Apple commercial-content requirements](#)

Read the Child Pricing and Monetization Plan, use both offline calculators, and implement section 17 plus acceptance tests M01-M20 and C01-C12. The package remains a specification; no campaigns or affiliate account have been activated.